

Physics

Time Allowed : 2 hour _____ **Maximum Marks : 200**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

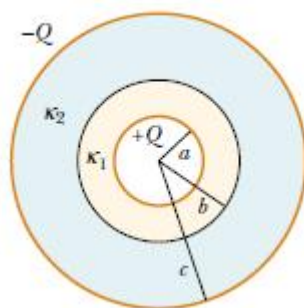
A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

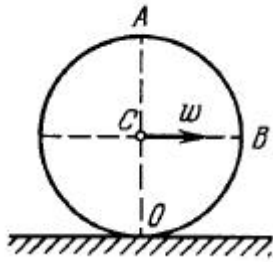
1. The question paper contains 20 questions. Each question carries 10 marks.

- Q1:** How would you determine the diameter of a molecule of oxalic acid?
- Q2:** A child loves to watch as you fill a transparent plastic bottle with shampoo. Every horizontal cross-section of the bottle is a circle, but the diameters of the circles all have different values, so that the bottle is much wider in some places than in others. You pour in bright green shampoo with constant volume flow rate $16.5 \text{ cm}^3/\text{s}$. At what rate is its level in the bottle rising (a) at a point where the diameter of the bottle is 6.30 cm and (b) at a point where the diameter is 1.35 cm?
- Q3:** Automotive engineers refer to the time rate of change of acceleration as the “jerk.” If an object moves in one dimension such that its jerk J is constant, (a) determine expressions for its acceleration a_x , velocity v_x , and position x , given that its initial acceleration, speed, and position are a_{xi} , v_{xi} , and x_i , respectively. (b) Show that $a_x^2 = a_{xi}^2 + 2J(v_x - v_{xi})$.
- Q4:** A conducting spherical shell has inner radius a and outer radius c . The space between these two surfaces is filled with a dielectric for which the dielectric constant is κ_1 between a and b , and κ_2 between b and c (Fig.). Determine the capacitance of this system.



- Q5:** Theory of relativity reveals that mass can be converted into energy. The energy E so obtained is proportional to certain powers of mass m and the speed c of light. Guess the relation among the quantities using the method of dimensions.
- Q6:** A radius vector of a point A relative to the origin varies with time t as $\mathbf{r} = at\hat{i} - bt^2\hat{j}$, where a and b are positive constants, and $\hat{i}$ and $\hat{j}$ are the unit vectors of the x and y axes. Find: (a) the equation of the point's trajectory $y(x)$; plot this function; (b) the time dependence of the velocity $\mathbf{v}$ and acceleration $\mathbf{w}$ vectors, as well as of the moduli of these quantities; (c) the time dependence of the angle α between the vectors $\mathbf{w}$ and $\mathbf{v}$; (d) the mean velocity vector averaged over the first t seconds of motion, and the modulus of this vector.
- Q7:** Two swimmers leave point A on one bank of the river to reach point B lying right across on the other bank. One of them crosses the river along the straight line AB while the other swims at right angles to the stream and then walks the distance that he has been carried away by the stream to get to point B . What was the velocity u of his walking if both swimmers reached the destination simultaneously? The stream velocity $v_o = 2.0 \text{ km/hour}$ and the velocity v' of each swimmer with respect to water equals 2.5 km per hour .
- Q8:** Point A moves uniformly with velocity $\mathbf{v}$ so that the vector $\mathbf{v}$ is continually "aimed" at point B which in its turn moves rectilinearly and uniformly with velocity $\mathbf{u} < \mathbf{v}$. At the initial moment of time $\mathbf{v} \perp \mathbf{u}$ and the points are separated by a distance l . How soon will the points converge?
- Q9:** A particle moves in the plane xy with constant acceleration $\mathbf{w}$ directed along the negative y axis. The equation of motion of the particle has the form $y = ax - bx^2$, where a and b are positive constants. Find the velocity of the particle at the origin of coordinates.
- Q10:** A ball of radius $R = 10.0 \text{ cm}$ rolls without slipping down an inclined plane so that its centre moves with constant acceleration u $w = 2.50 \text{ cm/s}^2$; $t = 2.00 \text{ s}$ after the beginning of motion its position

corresponds to that shown in Fig. . Find: (a) the velocities of the points A, B, and O; (b) the accelerations of these points.



- Q11:** What would be the interaction force between two copper spheres, each of mass 1 g, separated by the distance 1 m, if the total electronic charge in them differed from the total charge of the nuclei by one per cent?
- Q12:** A system consists of a thin charged wire ring of radius R and a very long uniformly charged thread oriented along the axis of the ring, with one of its ends coinciding with the centre of the ring. The total charge of the ring is equal to q . The charge of the thread (per unit length) is equal to λ . Find the interaction force between the ring and the thread.
- Q13:** An infinitely long cylindrical surface of circular crosssection is uniformly charged lengthwise with the surface density $\sigma = \sigma_0 \cos \phi$, where ϕ is the polar angle of the cylindrical coordinate system whose z axis coincides with the axis of the given surface. Find the magnitude and direction of the electric field strength vector on the z axis.
- Q14:** dipole with an electric moment p is located at a distance r from a long thread charged uniformly with a linear density A . Find the force F acting on the dipole if the vector p is oriented
- along the thread;
 - along the radius vector r ;
 - at right angles to the thread and the radius vector r .
- Q15:** The field potential in a certain region of space depends only on the x coordinate as $\phi = -ax^3 + b$, where a and b are constants. Find the distribution of the space charge $\rho(x)$.
- Q16:** Two point charges, q and $-q$, are separated by a distance l , both being located at a distance $l/2$ from the infinite conducting plane. Find:
- the modulus of the vector of the electric force acting on each charge;
 - the magnitude of the electric field strength vector at the midpoint between these charges.
- Q17:** A point charge q is located at a distance l from an infinite conducting plane. Determine the surface density of charges induced on the plane as a function of separation r from the base of the perpendicular drawn to the plane from the charge.
- Q18:** Two small identical balls carrying the charges of the same sign are suspended from the same point by insulating threads of equal length. When the surrounding space was filled with kerosene the divergence angle between the threads remained constant. What is the density of the material of which the balls are made?
- Q19:** A half-space filled with uniform isotropic dielectric with permittivity ϵ has the conducting boundary plane. Inside the dielectric, at a distance l from this plane, there is a small metal ball possessing a charge q . Find the surface density of the bound charges at the boundary plane as a function of distance r from the ball.
- Q20:** An infinitely long round dielectric cylinder is polarized uniformly and statically, the polarization P being perpendicular to the axis of the cylinder. Find the electric field strength E inside the dielectric.